NANJING UNIVERSITY

Laplacian Deformation

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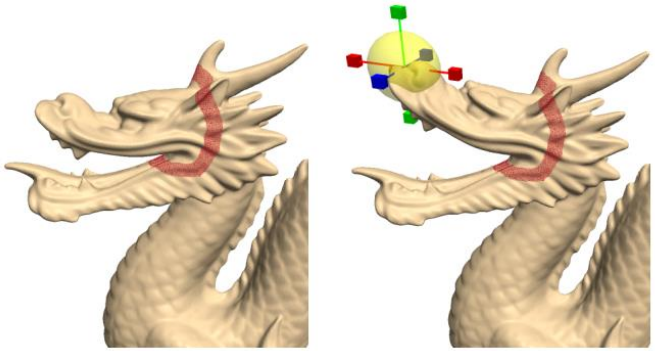
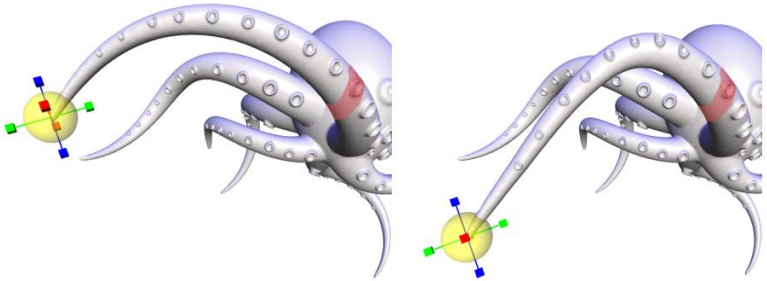
Outline

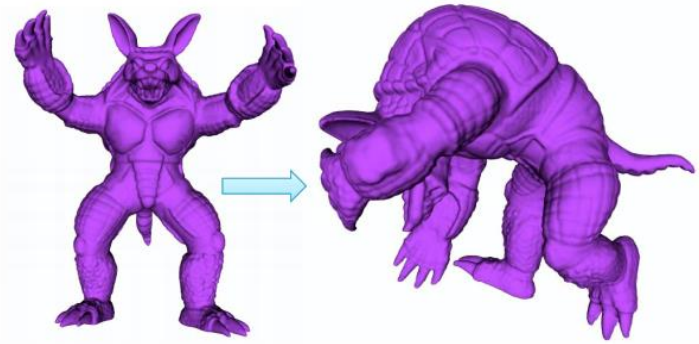
- Motivation and Application
- Laplacian Deformation
- Reference

Motivation & Application

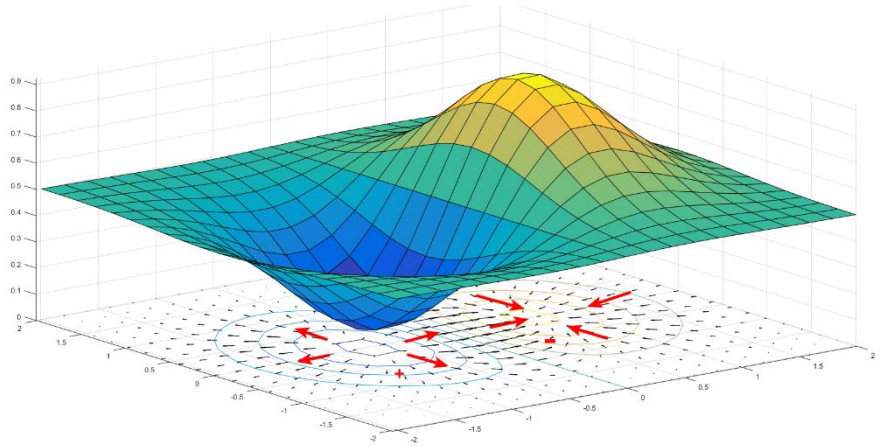
- Generate new model shapes by deforming existing ones







Surface Laplace Operator



$$\Delta f = \nabla^2 f = \nabla \bullet \nabla f$$

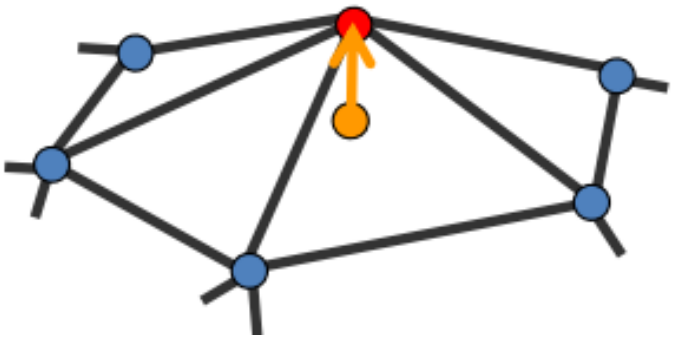
where $\nabla f = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}$

For 3-D European Space

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Discrete Laplace coordinates

For Mesh(V,E,F)



$$\delta_i = L(v_i) = v_i - \sum_{j \in N(i)} w_{ij} v_j$$

$\delta_i =$ Laplace coordinate of vi

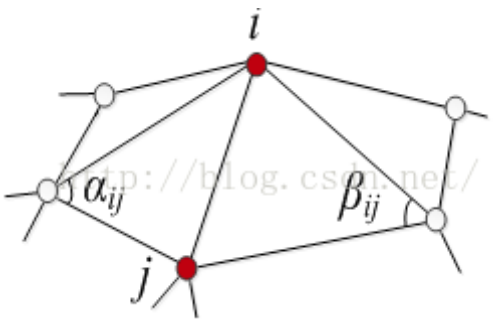
$L(\bullet) =$ Laplace operator of mesh

$w_{ij} =$ Weight of vi and vj

Weight

- Uniform weight

$$w_{ij} = \frac{1}{d_i}$$



- Coincident weight

$$w_{ij} = \frac{\cot \alpha_{ij} + \cot \beta_{ij}}{2}$$

Properties

- Linear transformation

$$\mathrm{T}(L(v_i)) = L(T(v_i))$$

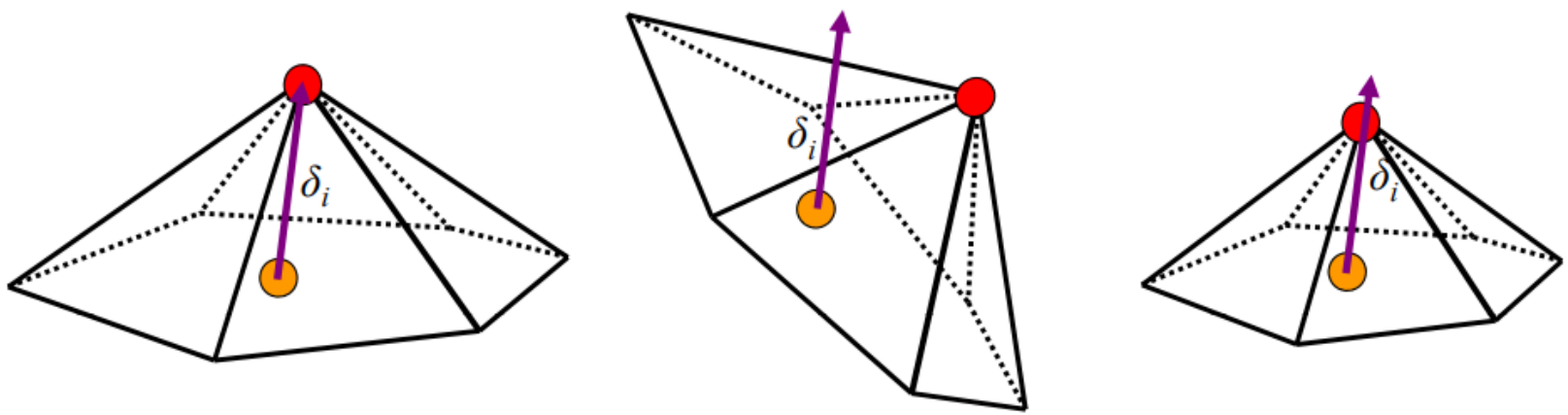
- Translation invariance

$$\sum_{j\in N(i)} \omega_{ij}(Tv_i - Tv_j) = \sum_{j\in N(i)} \omega_{ij} T(v_i - v_j) = \sum_{j\in N(i)} \omega_{ij} T \begin{pmatrix} x_{v_i} - x_{v_j} \\ y_{v_i} - y_{v_j} \\ z_{v_i} - z_{v_j} \\ 0 \end{pmatrix} = \sum_{j\in N(i)} \omega_{ij} \begin{pmatrix} x_{v_i} - x_{v_j} \\ y_{v_i} - y_{v_j} \\ z_{v_i} - z_{v_j} \\ 0 \end{pmatrix} = \delta_i$$

Properties

- Sensitive to Rotation/Scale

$$\sum_{j \in N(i)} \omega_{ij}(Rv_i - Rv_j) = R \sum_{j \in N(i)} \omega_{ij}(v_i - v_j) = R\delta_i$$



Laplacian Deformation

- For M(V,E)

$$LV = \delta$$

$$V = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_i \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_i & y_i & z_i \\ \vdots & \vdots & \vdots \\ x_n & y_n & z_n \end{bmatrix}, \quad \delta = \begin{bmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_i \\ \vdots \\ \delta_n \end{bmatrix} = \begin{bmatrix} \delta_{1x} & \delta_{1y} & \delta_{1z} \\ \delta_{2x} & \delta_{2y} & \delta_{2z} \\ \vdots & \vdots & \vdots \\ \delta_{ix} & \delta_{iy} & \delta_{iz} \\ \vdots & \vdots & \vdots \\ \delta_{nx} & \delta_{ny} & \delta_{nz} \end{bmatrix}$$

http://blog.csdn.net/

$$L_{ij} = \begin{cases} 1 \quad , \quad i = j \\ -\omega_{ij}, (i,j) \in E \\ 0 \quad , \quad \text{others} \end{cases}$$

Laplacian Matrix

- Pose modeling constraints for vertices $C \subset V$
  - $v'_i = u_i \quad i \in C$

- No exact solution, minimize error

$$\mathbf{V}' = \arg\min_{\mathbf{V}'} \sum_{i=1}^{n} \|\delta_i - L(\mathbf{v}'_i)\|^2 + \sum_{i \in C} \|\mathbf{v}'_i - \mathbf{u}_i\|^2$$

Laplacian coordinates of original mesh

Laplacian coordinates of deformed mesh

User Constraints

Linear Least Squares

- $\boldsymbol{Ax} = \boldsymbol{b}$ with m equations, n unknowns

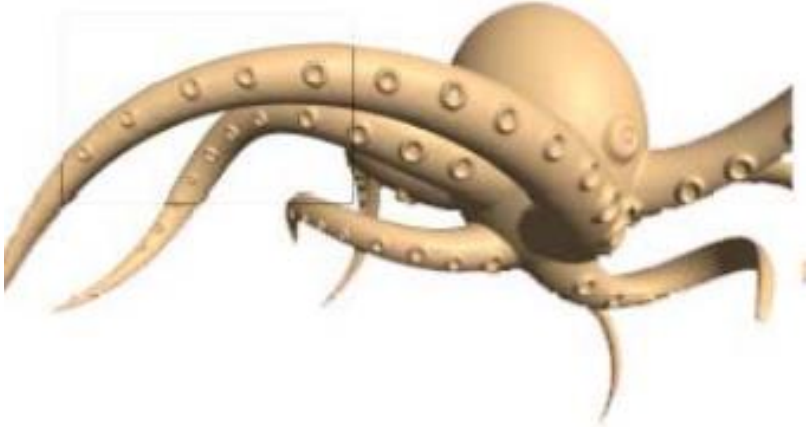
- Normal equations: $(A^{\mathrm{T}}A)\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$
- Solution by pseudo inverse:

$$\boldsymbol{x} = A^{+}\boldsymbol{b} = [(A^{T}A)^{-1}A^{\mathrm{T}}]\boldsymbol{b}$$

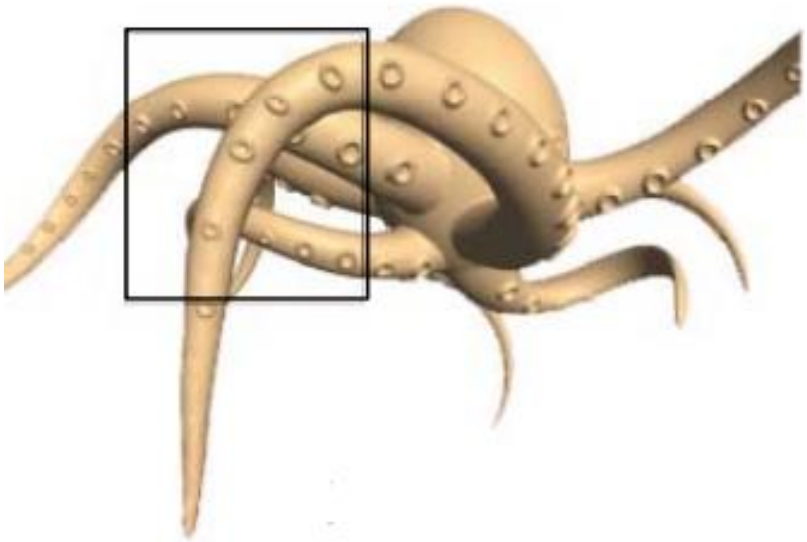
If system under determined: $\boldsymbol{x} = \boldsymbol{arg\ min}\ \{\ \|\boldsymbol{x}\| : A\boldsymbol{x} = \boldsymbol{b}\ \}$

If system over determined: $\boldsymbol{x} = \boldsymbol{arg\ min}\ \{\ \|A\boldsymbol{x}-\boldsymbol{b}\|^2\ \}$

Result



Origin Model



Deform Model

Reference

- Sorkine O, Cohen-Or D, Lipman Y, et al. Laplacian surface editing[C]//Proceedings of the 2004 Eurographics/ACM SIGGRAPH symposium on Geometry processing. ACM, 2004: 175-184.

Oneline Demo

http://fanghao.site/OnlineDemos/Laplace%20Deform/demo.html